Sensitivity report: p3-morris-capacity

morris · 1180 runs · 2026-11-09 11:46 UTC

- Screening summary
- throughput_veh_h
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Morris elementary-effects screening (G§8.4) over 58 factors, 10 trajectories on a 4-level grid (590 points × replications). μ* ranks influence; σ above the diagonal σ = μ* marks non-linear or interacting effects. Effects are per unit of each factor (register unit; the mean for distributions, ADR-0006); `mu_star_rel` compares factors within one output.

Screening summary

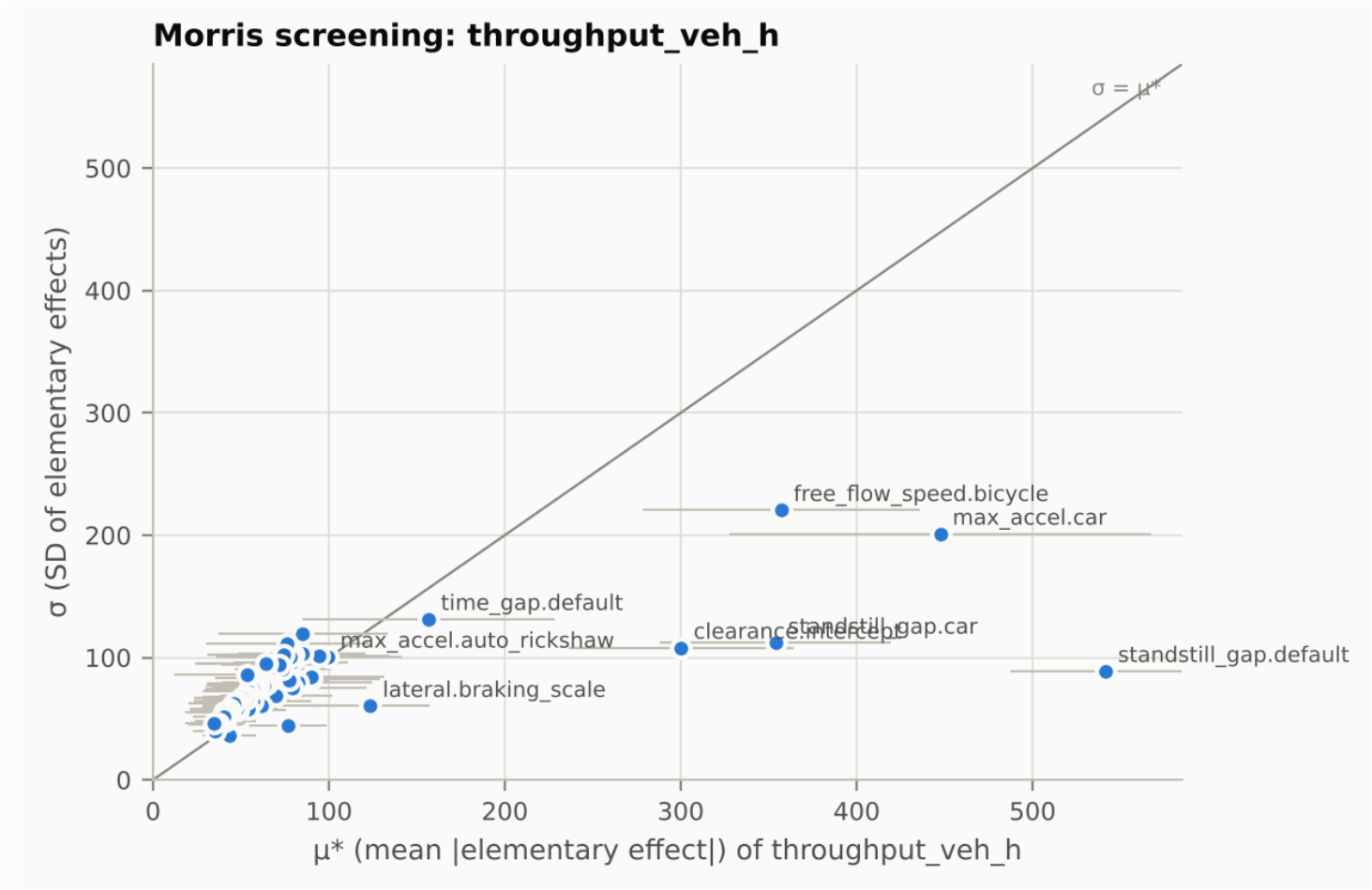
Relative μ* (μ* over the largest μ* of each output). 0 of 58 factors stay below 5% of the most influential factor for every output: candidates to hold fixed in the next stage. The cut-off is a reporting convention for this screen, not a model parameter.

Relative μ* by output

factor	delay_mean_s	delay_p95_s	network_mean_speed_m_s	throughput_veh_h	max_rel
vehicles.free_flow_speed.bicycle	1	1	1	0.6595	1
interaction.following.pair_multiplier.standstill_gap.default	0.143	0.1115	0.1239	1	1
interaction.following.max_accel.car	0.2066	0.1721	0.2037	0.8263	0.8263
interaction.following.standstill_gap.car	0.08176	0.09731	0.07105	0.6536	0.6536
interaction.lateral.clearance.intercept	0.09958	0.1106	0.115	0.5544	0.5544
interaction.following.pair_multiplier.time_gap.default	0.06097	0.07599	0.0724	0.2892	0.2892
interaction.lateral.braking_scale	0.08373	0.158	0.0853	0.2282	0.2282
interaction.following.max_accel.auto_rickshaw	0.08333	0.08247	0.06499	0.1835	0.1835
interaction.following.comfortable_decel.car	0.07317	0.07268	0.05776	0.1742	0.1742
vehicles.free_flow_speed.motorcycle	0.1185	0.1434	0.1679	0.08958	0.1679
interaction.following.max_accel.two_wheeler	0.08531	0.09547	0.1004	0.1664	0.1664
interaction.following.time_gap.car	0.08353	0.1083	0.07764	0.1659	0.1659
vehicles.free_flow_speed.truck	0.06967	0.09422	0.08081	0.1571	0.1571
vehicles.free_flow_speed.sedan	0.08538	0.1147	0.09683	0.1562	0.1562
interaction.following.standstill_gap.two_wheeler	0.07921	0.09223	0.0769	0.1516	0.1516
interaction.following.standstill_gap.e_rickshaw	0.08408	0.09489	0.05902	0.1472	0.1472
vehicles.free_flow_speed.e_rickshaw	0.1045	0.04631	0.1142	0.1459	0.1459
demand.min_headway.car	0.07969	0.1044	0.08773	0.1449	0.1449
vehicles.free_flow_speed.auto_rickshaw	0.1016	0.1277	0.1444	0.1193	0.1444
interaction.following.max_accel.lcv	0.04841	0.09632	0.04538	0.1443	0.1443
interaction.following.time_gap.e_rickshaw	0.07509	0.09501	0.05742	0.1426	0.1426
interaction.following.standstill_gap.auto_rickshaw	0.06172	0.06829	0.05322	0.1418	0.1418
interaction.following.time_gap.two_wheeler	0.06803	0.08617	0.07155	0.1409	0.1409
interaction.following.time_gap.bus	0.1085	0.08934	0.1064	0.1366	0.1366
vehicles.free_flow_speed.moped	0.06175	0.08786	0.05435	0.1364	0.1364
interaction.following.comfortable_decel.e_rickshaw	0.08873	0.1249	0.07708	0.1325	0.1325
interaction.following.standstill_gap.bus	0.08745	0.1096	0.09815	0.1293	0.1293
interaction.following.time_gap.truck	0.0772	0.1288	0.06765	0.08941	0.1288
vehicles.free_flow_speed.lcv	0.1067	0.08984	0.1059	0.1212	0.1212

factor	delay_mean_s	delay_p95_s	network_mean_speed_m_s	throughput_veh_h	max_rel
vehicles.free_flow_speed.hatchback	0.09126	0.1104	0.08294	0.1193	0.1193
demand.min_headway.two_wheeler	0.08854	0.07226	0.07589	0.119	0.119
interaction.following.comfortable_decel.bicycle	0.05321	0.08586	0.06174	0.1165	0.1165
interaction.following.standstill_gap.lcv	0.07743	0.1145	0.0527	0.06797	0.1145
interaction.lateral.clearance.slope	0.06639	0.09423	0.07607	0.1138	0.1138
demand.min_headway.bicycle	0.07024	0.1131	0.05135	0.08975	0.1131
demand.min_headway.lcv	0.06441	0.1122	0.06123	0.07961	0.1122
demand.min_headway.bus	0.0579	0.1113	0.07166	0.07795	0.1113
interaction.following.comfortable_decel.two_wheeler	0.06252	0.09666	0.06877	0.1099	0.1099
interaction.following.time_gap.lcv	0.04459	0.05462	0.05257	0.1099	0.1099
interaction.following.comfortable_decel.lcv	0.0793	0.1073	0.06617	0.1045	0.1073
interaction.following.max_accel.bicycle	0.03787	0.06192	0.03954	0.1065	0.1065
interaction.following.comfortable_decel.auto_rickshaw	0.05061	0.07106	0.05025	0.1049	0.1049
vehicles.free_flow_speed.scooter	0.06057	0.09379	0.06805	0.1	0.1
interaction.following.time_gap.bicycle	0.07234	0.09551	0.08018	0.09988	0.09988
vehicles.free_flow_speed.bus	0.08945	0.05603	0.09923	0.09855	0.09923
interaction.following.comfortable_decel.bus	0.04419	0.09884	0.04921	0.09274	0.09884
vehicles.median_band.speed_bonus	0.02815	0.06015	0.0424	0.09872	0.09872
vehicles.free_flow_speed.suv	0.04626	0.04738	0.04527	0.09756	0.09756
interaction.following.max_accel.truck	0.0553	0.07898	0.05387	0.08792	0.08792
demand.min_headway.auto_rickshaw	0.06891	0.06978	0.06718	0.08509	0.08509
interaction.following.standstill_gap.truck	0.07282	0.08471	0.06739	0.0806	0.08471
interaction.following.max_accel.bus	0.08174	0.07875	0.07998	0.08011	0.08174
demand.min_headway.e_rickshaw	0.0396	0.05375	0.03017	0.07595	0.07595
interaction.following.max_accel.e_rickshaw	0.05092	0.0545	0.03882	0.07479	0.07479
interaction.following.comfortable_decel.truck	0.04023	0.05118	0.04976	0.07462	0.07462
interaction.following.time_gap.auto_rickshaw	0.05119	0.0732	0.0469	0.06548	0.0732
demand.min_headway.truck	0.05458	0.06581	0.06877	0.06565	0.06877
interaction.following.standstill_gap.bicycle	0.04534	0.04367	0.0378	0.06415	0.06415

throughput_veh_h



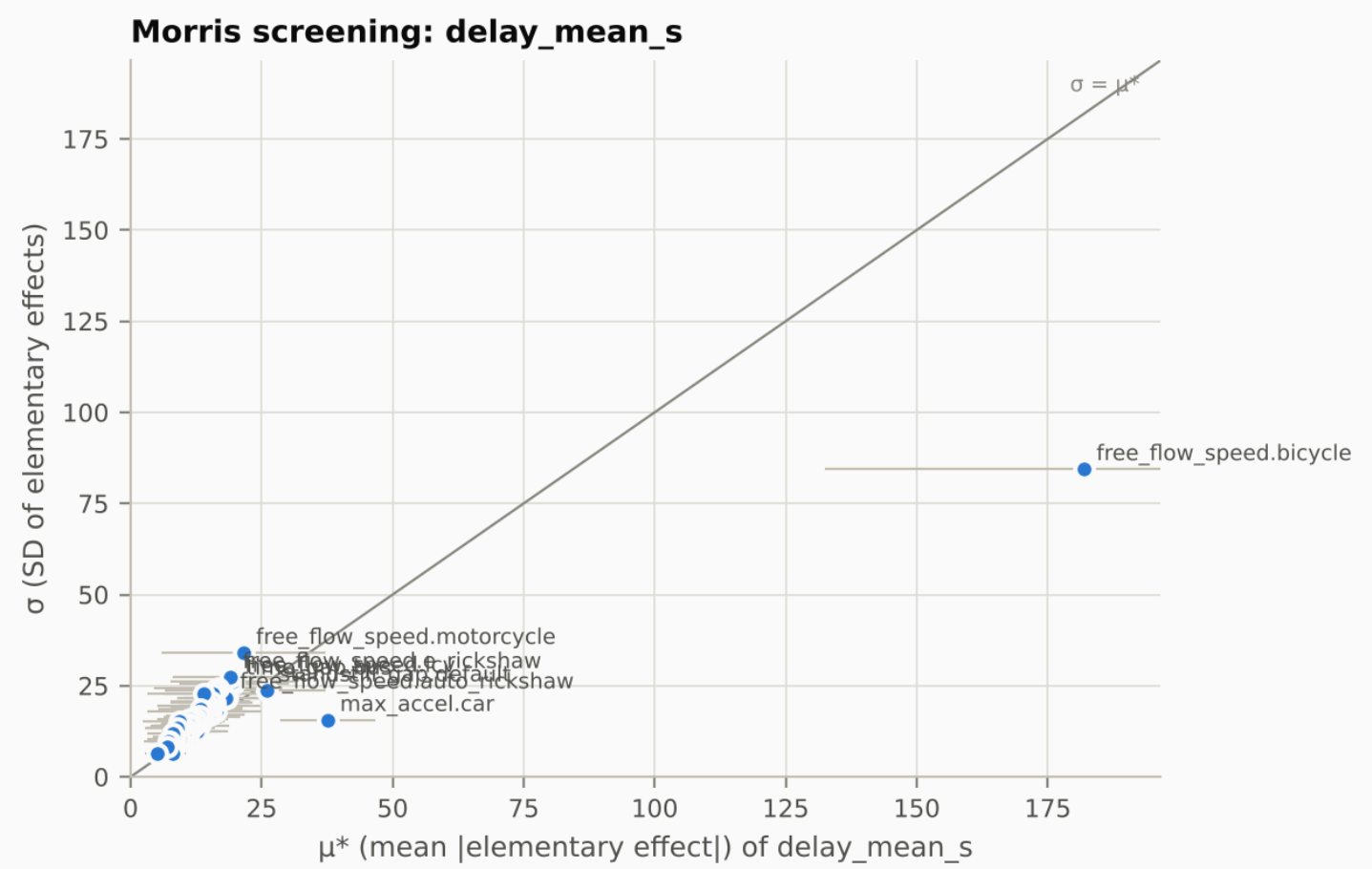
Morris μ*–σ for throughput_veh_h

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel
interaction.following.pair_multiplier.standstill_gap.default	-541.5	541.5	53.83	88.81	1
interaction.following.max_accel.car	447.5	447.5	120	201.1	0.8263
vehicles.free_flow_speed.bicycle	316.4	357.1	78.77	220.4	0.6595
interaction.following.standstill_gap.car	-354	354	65.71	111.9	0.6536
interaction.lateral.clearance.intercept	-300.2	300.2	63.93	107.9	0.5544
interaction.following.pair_multiplier.time_gap.default	-148.1	156.6	71.4	131.3	0.2892
interaction.lateral.braking_scale	-123.6	123.6	33.66	60.29	0.2282
interaction.following.max_accel.auto_rickshaw	72.36	99.36	41.96	100.5	0.1835
interaction.following.comfortable_decel.car	-62.28	94.32	39.99	101.4	0.1742
interaction.following.max_accel.two_wheeler	77.49	90.09	38.62	82.31	0.1664
interaction.following.time_gap.car	-77.76	89.82	41.7	84.42	0.1659
vehicles.free_flow_speed.truck	13.59	85.05	48.07	119.7	0.1571
vehicles.free_flow_speed.sedan	30.96	84.6	36.02	103.3	0.1562
interaction.following.standstill_gap.two_wheeler	-75.24	82.08	42.37	80.05	0.1516
interaction.following.standstill_gap.e_rickshaw	18.36	79.74	36.46	101.9	0.1472
vehicles.free_flow_speed.e_rickshaw	-74.88	79.02	42.65	75.49	0.1459
demand.min_headway.car	26.46	78.48	32.34	96.09	0.1449
interaction.following.max_accel.lcv	40.86	78.12	42.34	100.7	0.1443
interaction.following.time_gap.e_rickshaw	-37.62	77.22	23.47	81.36	0.1426
interaction.following.standstill_gap.auto_rickshaw	-74.79	76.77	22.29	43.97	0.1418
interaction.following.time_gap.two_wheeler	-12.6	76.32	46.11	111.1	0.1409
interaction.following.time_gap.bus	-25.02	73.98	43.2	102.4	0.1366

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel
vehicles.free_flow_speed.moped	-4.95	73.89	27.29	90.96	0.1364
interaction.following.comfortable_decel.e_rickshaw	-9.45	71.73	32.81	94.52	0.1325
interaction.following.standstill_gap.bus	-57.96	70.02	31.73	68.62	0.1293

33 more rows in `tables/morris_throughput_veh_h.csv`.

delay_mean_s



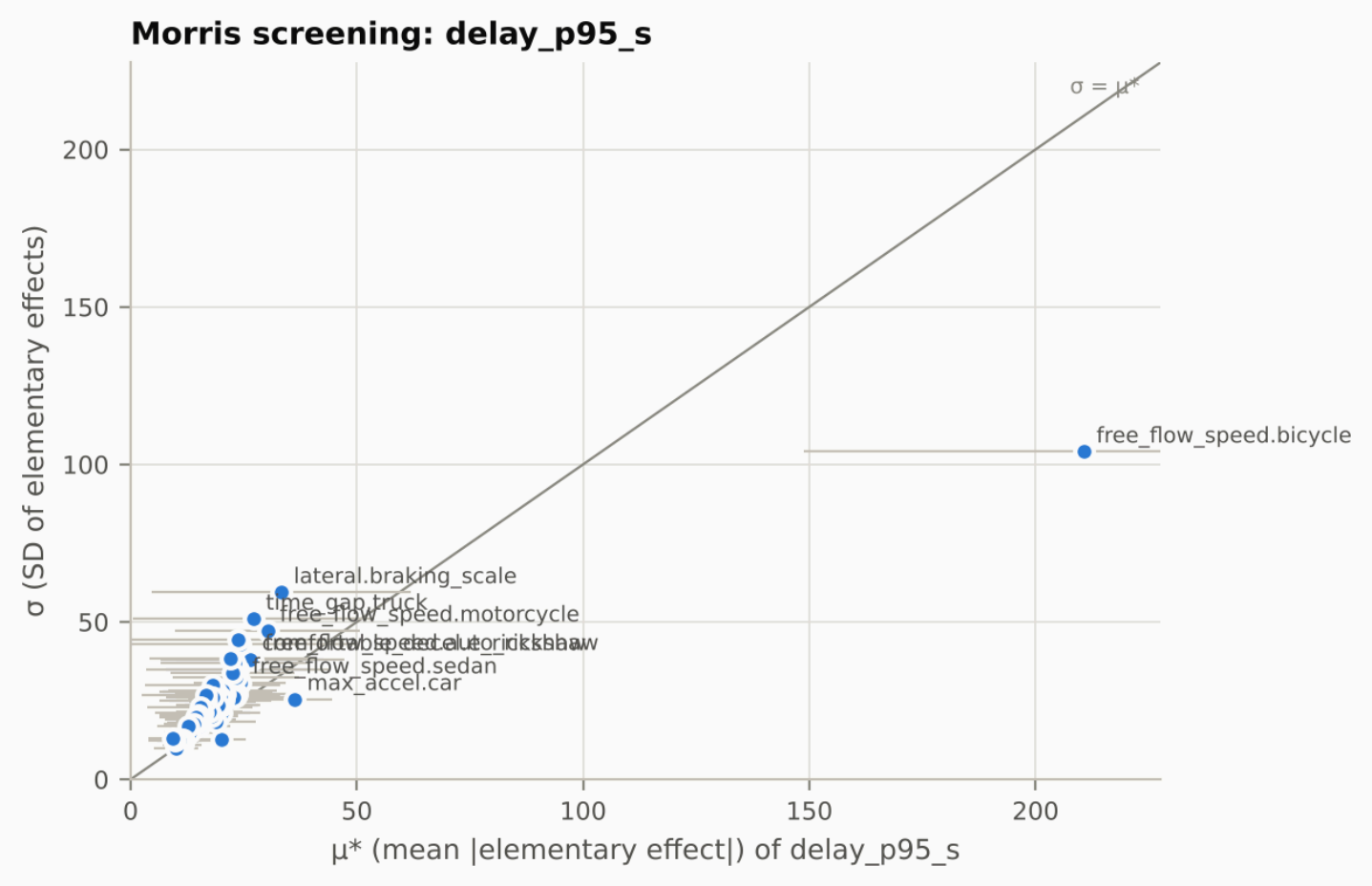
Morris μ*–σ for delay_mean_s

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel
vehicles.free_flow_speed.bicycle	-181.9	181.9	49.36	84.33	1
interaction.following.max_accel.car	37.59	37.59	9.022	15.52	0.2066
interaction.following.pair_multiplier.standstill_gap.default	-22.05	26.01	11.09	23.68	0.143
vehicles.free_flow_speed.motorcycle	-9.832	21.55	15.64	33.94	0.1185
interaction.following.time_gap.bus	9.185	19.75	10.42	25.55	0.1085
vehicles.free_flow_speed.lcv	-10.85	19.42	11.79	26.2	0.1067
vehicles.free_flow_speed.e_rickshaw	0.5953	19.01	10.94	27.31	0.1045
vehicles.free_flow_speed.auto_rickshaw	-9.697	18.49	8.263	21.74	0.1016
interaction.lateral.clearance.intercept	-11.6	18.12	9.202	21.56	0.09958
vehicles.free_flow_speed.hatchback	10.4	16.6	12.24	24.34	0.09126
vehicles.free_flow_speed.bus	-7.865	16.27	5.939	17.86	0.08945
interaction.following.comfortable_decel.e_rickshaw	-1.183	16.14	9.494	23.75	0.08873
demand.min_headway.two_wheeler	-2.227	16.11	6.666	20.3	0.08854
interaction.following.standstill_gap.bus	-1.024	15.91	6.818	20.48	0.08745

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel
vehicles.free_flow_speed.sedan	-2.616	15.53	9.309	22.73	0.08538
interaction.following.max_accel.two_wheeler	-6.333	15.52	7.807	20.36	0.08531
interaction.following.standstill_gap.e_rickshaw	2.574	15.3	5.516	18.44	0.08408
interaction.lateral.braking_scale	-1.98	15.23	7.66	20.66	0.08373
interaction.following.time_gap.car	-1.826	15.2	5.211	18.2	0.08353
interaction.following.max_accel.auto_rickshaw	6.097	15.16	5.106	17.13	0.08333
interaction.following.standstill_gap.car	-8.981	14.87	6.806	17.11	0.08176
interaction.following.max_accel.bus	12.53	14.87	9.848	19.46	0.08174
demand.min_headway.car	-1.808	14.5	6.46	18.78	0.07969
interaction.following.comfortable_decel.lcv	-3.565	14.43	6.869	18.9	0.0793
interaction.following.standstill_gap.two_wheeler	0.5661	14.41	6.15	18.72	0.07921

33 more rows in `tables/morris_delay_mean_s.csv`.

delay_p95_s



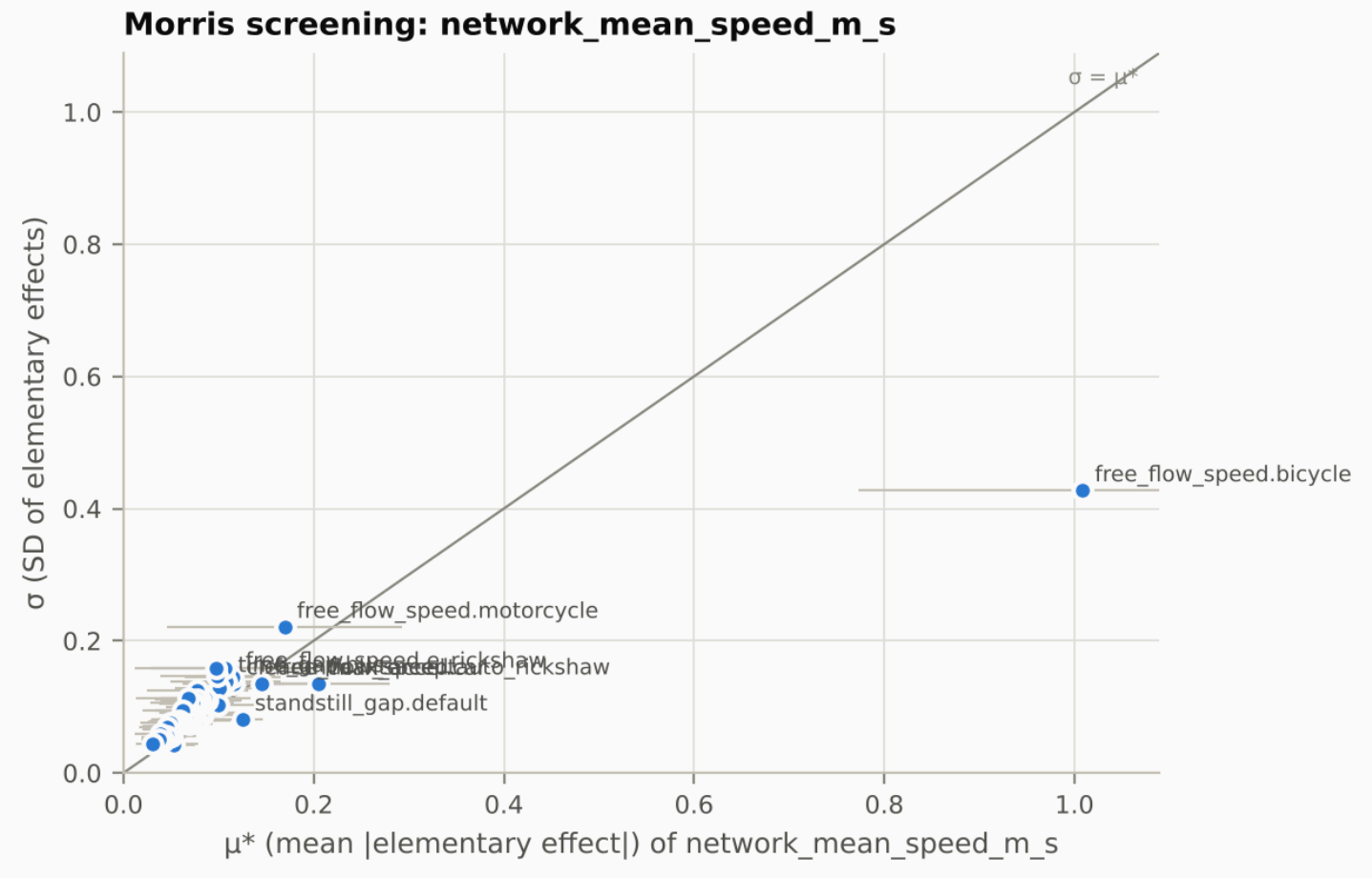
Morris μ*–σ for delay_p95_s

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel
vehicles.free_flow_speed.bicycle	-210.7	210.7	61.88	104	1
interaction.following.max_accel.car	30.35	36.26	8.292	25.43	0.1721
interaction.lateral.braking_scale	5.962	33.29	28.63	59.37	0.158
vehicles.free_flow_speed.motorcycle	8.429	30.23	20.38	47.24	0.1434
interaction.following.time_gap.truck	18.67	27.14	26.65	50.95	0.1288
vehicles.free_flow_speed.auto_rickshaw	-21.47	26.9	20.28	38.18	0.1277

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel
interaction.following.comfortable_decel.e_rickshaw	-10.07	26.31	16.16	38.1	0.1249
vehicles.free_flow_speed.sedan	3.932	24.16	10.19	30.76	0.1147
interaction.following.standstill_gap.lcv	-21.88	24.12	25.62	42.98	0.1145
demand.min_headway.bicycle	-20.36	23.82	20.28	34.92	0.1131
demand.min_headway.lcv	-6.951	23.65	23.55	44.46	0.1122
interaction.following.pair_multiplier.standstill_gap.default	-8.388	23.5	15.64	34.81	0.1115
demand.min_headway.bus	-17.74	23.45	12.33	27.26	0.1113
interaction.lateral.clearance.intercept	-12.64	23.3	9.687	26.43	0.1106
vehicles.free_flow_speed.hatchback	8.992	23.26	13.86	32.56	0.1104
interaction.following.standstill_gap.bus	-5.953	23.09	16.57	36.81	0.1096
interaction.following.time_gap.car	-17.38	22.81	12.66	26.18	0.1083
interaction.following.comfortable_decel.lcv	3.776	22.61	13.72	33.78	0.1073
demand.min_headway.car	-7.007	22	17.8	38.23	0.1044
interaction.following.comfortable_decel.bus	-9.365	20.83	11.01	27.21	0.09884
interaction.following.standstill_gap.car	1.953	20.5	7.786	25.39	0.09731
interaction.following.comfortable_decel.two_wheeler	-0.1863	20.37	7.316	24.79	0.09666
interaction.following.max_accel.lcv	9.103	20.3	11.96	28.01	0.09632
interaction.following.time_gap.bicycle	8.017	20.13	8.026	23.74	0.09551
interaction.following.max_accel.two_wheeler	-13.96	20.12	8.566	21.02	0.09547

33 more rows in `tables/morris_delay_p95_s.csv`.

network_mean_speed_m_s



Morris μ*–σ for network_mean_speed_m_s

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel
vehicles.free_flow_speed.bicycle	0.9992	1.008	0.2352	0.4272	1
interaction.following.max_accel.car	-0.2054	0.2054	0.07511	0.1346	0.2037
vehicles.free_flow_speed.motorcycle	0.1558	0.1693	0.1238	0.2207	0.1679
vehicles.free_flow_speed.auto_rickshaw	0.1352	0.1456	0.07324	0.1352	0.1444
interaction.following.pair_multiplier.standstill_gap.default	0.1049	0.1249	0.02134	0.08052	0.1239
interaction.lateral.clearance.intercept	0.0685	0.116	0.05396	0.1351	0.115
vehicles.free_flow_speed.e_rickshaw	0.03445	0.1152	0.05027	0.1452	0.1142
interaction.following.time_gap.bus	-0.05229	0.1073	0.05757	0.1387	0.1064
vehicles.free_flow_speed.lcv	0.0631	0.1068	0.07695	0.1583	0.1059
interaction.following.max_accel.two_wheeler	0.045	0.1013	0.047	0.1274	0.1004
vehicles.free_flow_speed.bus	0.06184	0.1001	0.03706	0.1035	0.09923
interaction.following.standstill_gap.bus	0.02439	0.09897	0.06008	0.1463	0.09815
vehicles.free_flow_speed.sedan	0.07978	0.09763	0.08547	0.1583	0.09683
demand.min_headway.car	0.002151	0.08846	0.03363	0.1093	0.08773
interaction.lateral.braking_scale	0.04474	0.08601	0.04759	0.1136	0.0853
vehicles.free_flow_speed.hatchback	0.06024	0.08363	0.03602	0.08719	0.08294
vehicles.free_flow_speed.truck	0.02576	0.08149	0.04228	0.1088	0.08081
interaction.following.time_gap.bicycle	-0.01127	0.08085	0.02991	0.09925	0.08018
interaction.following.max_accel.bus	-0.06183	0.08065	0.04122	0.0915	0.07998
interaction.following.time_gap.car	-0.01377	0.07829	0.03339	0.09985	0.07764
interaction.following.comfortable_decel.e_rickshaw	0.01513	0.07772	0.05319	0.124	0.07708
interaction.following.standstill_gap.two_wheeler	0.01952	0.07754	0.0449	0.1116	0.0769
interaction.lateral.clearance.slope	0.05834	0.07671	0.03546	0.07928	0.07607
demand.min_headway.two_wheeler	0.01039	0.07652	0.03919	0.1047	0.07589
interaction.following.pair_multiplier.time_gap.default	-0.03465	0.073	0.03515	0.09103	0.0724

33 more rows in `tables/morris_network_mean_speed_m_s.csv`.